Amirali Imanipanah

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Professional Summary

Final-year Electrical Engineering student specializing in **Digital Electronics**, **FPGA-based Design**, and **Embedded Systems**. Proficient in high-speed **PCB Design** and autonomous robotics using **ROS2**. Experienced in implementing high-frequency hardware algorithms and integrating AI with classical control systems. Proven leadership through multiple Teaching Assistantships in microcontrollers and electronics.

Education

University of Isfahan — Isfahan, Iran
Bachelor of Science in Electrical Engineering - Electronics — Sep 2022 – July 2026 (Expected)

- **Overall GPA:** 16.50 / 20.00
- **Key Courses:** Microcontrollers, Digital Systems Design, Computer Architecture, Signals & Systems.

Research Interests

Digital Systems: FPGA-based design, Embedded Linux, Verilog/VHDL, and SoC.

Robotics: Autonomous Navigation, ROS2, and Real-time Systems.

AI in Hardware: Machine Learning on Microcontrollers (TinyML) and Computer Vision.

Technical Skills

Embedded & Robotics: STM32/STM8, RP2040, AVR, Raspberry Pi 5 (Embedded Linux), ROS2, ESP32, IOT Protocols.

Hardware Design: Altium Designer (Advanced), KiCad, Proteus, PSpice, SMD/DIP Soldering & Assembly.

Programming & Tools: C/C++, Verilog, Python, MATLAB, HTML/CSS/JS, Git, Linux CLI, SolidWorks.

Certifications

Professional PCB Design (Advanced level): Issued by IEEE Student Branch. — 2024

Introductory PCB Design (Altium Designer): Issued by IEEE Student Branch. — 2023

Teaching Experience

Teaching Assistant (TA) — *University of Isfahan*

- **Microcontrollers (Spring 2026)**: Guided 80+ students in AVR/STM32 interfacing and programming.
- **Electronics I (Spring 2026)**: Conducted recitation sessions and assisted with transistor circuits.
- **Electric Circuits I (Fall 2025)**: Conducted problem-solving workshops and network analysis tutorials.
- **Fundamentals of Programming (Fall 2025)**: Taught C++ logic, structural debugging, and basic algorithms.

Selected Projects

Autonomous Mobile Robot with AI-Optimized Control — *B.Sc. Thesis*

- Developed a ROS2/Raspberry Pi 5 robot integrating **SLAM**, **Nav2**, and edge AI for autonomous navigation.
- Implemented a **Neural Network-optimized PID controller** to solve dynamic stability, slope, and friction challenges.
- Designed custom power distribution and motor control PCBs utilizing Altium Designer.

High-Frequency CORDIC Processor Core — *Digital Systems Design Project*

- Implemented a pipelined CORDIC algorithm on FPGA for hardware-level trigonometric operations, achieving a maximum operating frequency of **115.399 MHz**.

High-Performance STM32-Based Graphical HMI System — *Independent Hardware Project*

- Designed an embedded system using an **STM32F407** to drive a 40-pin TFT LCD and capacitive touch panel.
- Developed a mathematical core for real-time graphing and matrix operations, suitable for industrial HMI use.

Analysis of MQ-Series Gas Sensors — *Electronic Measurement Course*

- Investigated the electrochemical operation and internal electrode structures of MQ-family sensors.

Languages

English: Professional working proficiency (Preparing for IELTS). **Persian:** Native.